Examining Microlensing Data

1. Abstract

a. In this report, I compare flux against time in order to make a light curve. My figures analyze flux over time, and then I fit a microlensing model to one of the files.

2. Equations and Data

a. I looked at MOA data from the nasa exoplanet archive(*MOA Mission and Dataset Information*, n.d.), and fit it using microlensing equations

b. The microlensing equations (Mróz et al., 2024):

c.

$$A_{\mathrm{PSPL}}(t) = \frac{u(t)^2 + 2}{u(t)\sqrt{u(t)^2 + 4}}$$

$$u(t) \;=\; \sqrt{\left(\left(t - t_0\right)/t_{\mathrm{E}}\right)^2 + u_0^2}$$

$$F(t) = F_0\left[1 + f_{\mathrm{s}}(A_{\mathrm{PSPL}}(t) - 1)\right]$$

d. Variables:

   i. Fs is the blending parameter
   ii. t is time
   iii. Fb is the baseline flux
   iv. uo is the impact parameter
   v. tE is the Einstein radius crossing timescale

3. Graphing

a. I first had to open the file, which sounded easy but I was unfamiliar with the .tar datatype, so I imported tarfile, and gzip to help get to the real data. I also imported Table from astropy.table and curve_fit from scipy.optimize. In order to handle and manipulate the data I used numpy and pandas, and matplotlib for graphing the values.

b. Then I cleaned the data, and the only things I wanted to graph were flux and HJD. These values correspond to brightness and the date of the observational epoch (from the perspective of the sun).

c. I originally picked a random file to graph, but then I decided I wanted a more comprehensive understanding. I made my original code that graphed the file into a function and then I graphed the original random file, and found files that had less extreme points and more clear peaks to make more figures.

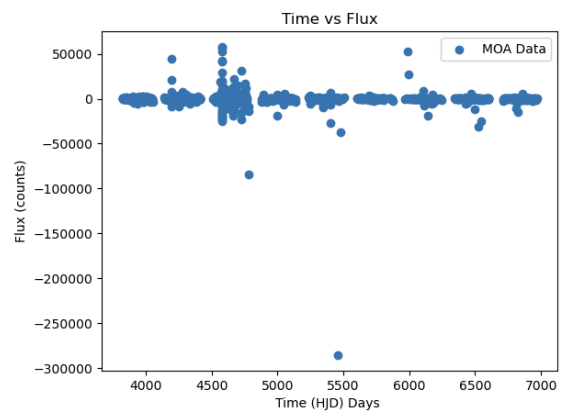
d. I then fit one of the models with the microlensing model

4. Analysis

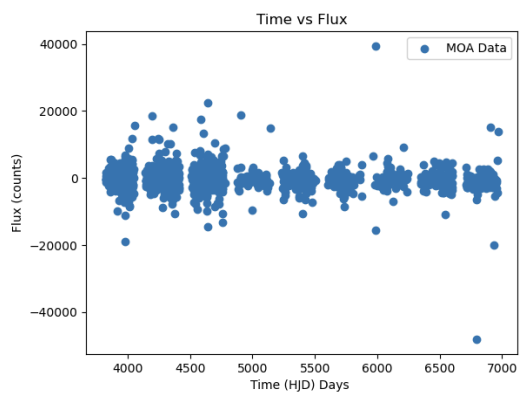
a. The consistent gaps in the data were confusing at first, but it appears to just be the time periods where there was no data collected by the telescope.
b. Graphing the flux over time allows us to see if there is a microlensing event, because there would be a spike in the flux. All of these graphs are relatively uniform, so they are very unlikely to microlensing events. I tried to look for files more likely to be said to be events by assigning them a score and graphing the file with the best score, but then I found a file that was dominated by a strong outlier. I then filtered out data with values over five standard deviations away.
c. I chose the most interesting graph to fit the microlensing model above.

5. Results

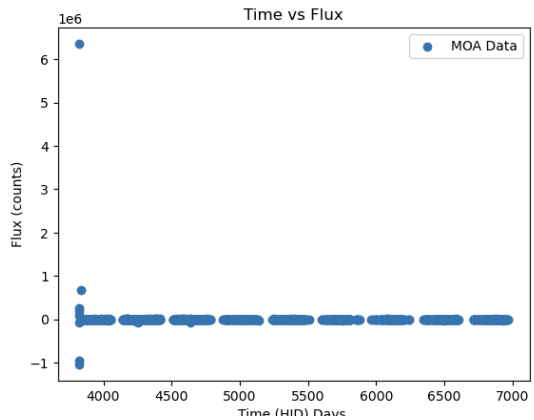
Randomly selected file:



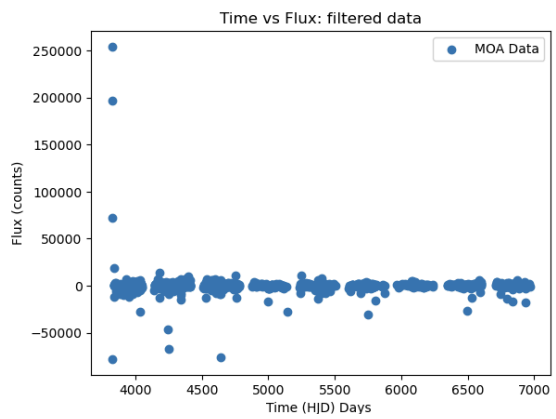
Second Randomly selected file:



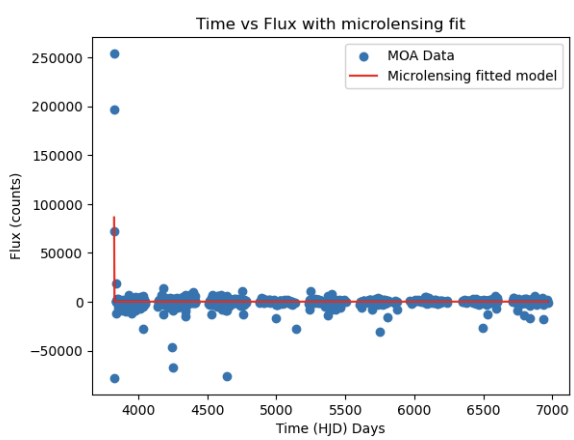
File found by looking for the best peaks:



File found by filtering out extreme values:



File fit by microlensing model:



Bibliography

MOA Mission and Dataset Information. (n.d.). Retrieved April 26, 2026, from https://exoplanetarchive.ipac.caltech.edu/docs/MOAMission.html?utm

Mróz, P., Udalski, A., Szymański, M. K., Soszyński, I., Pietrukowicz, P., Kozłowski, S., Poleski, R., Skowron, J., Ulaczyk, K., Gromadzki, M., Rybicki, K., Iwanek, P., Wrona, M., & Mróz, M. J. (2024). Limits on planetary-mass primordial black holes from the OGLE high-cadence survey of the Magellanic Clouds. *The Astrophysical Journal Letters*, *976*(1), L19. https://doi.org/10.3847/2041-8213/ad8e68